

Capstone Project: Checkout Experiment & Growth Analytics

Part A — Problem Framing Document

1. Business Objective

To determine if **Variant B** (the new checkout experience) creates a statistically significant uplift in purchase completion and revenue per session, while maintaining or improving margins and technical stability. The final analysis will provide a data-backed recommendation on whether to roll out Variant B to 100% of eligible users.

2. North Star Metric

Checkout Completion Rate (CCR)

$CCR = \text{Total Successful Purchases} / \text{Sessions with Begin Checkout}$

- **Rationale:** This metric isolates the checkout experience. By filtering for users who have already reached the "Begin Checkout" stage, we remove top-of-funnel noise and measure the specific efficacy of the new UI.

3. Supporting KPIs

KPI Category	KPI Name	Definition	Business Purpose
Primary	Session Conversion Rate (CVR)	Total Purchases / Total Eligible Sessions	High-level success metric for the entire session.
Financial	Revenue per Eligible Session (RPES)	Net Revenue / Total Eligible Sessions	The "Bottom Line" used for 30-day impact forecasting.
Basket	Average Order Value (AOV)	Total Net Revenue / Total Orders	Ensures Variant B doesn't lower total spend per user.
Efficiency	Funnel Step Conversions	Number of users in Step n+ 1 / Number of users in Steps n	Identifies exact page leaks (Shipping, Payment, etc.).
Technical	Payment Failure Rate	Failed Payment Events / Total Payment Attempts	Guardrail metric to detect technical bugs in Variant B.
Profitability	Gross Margin Proxy	(Net Revenue - Total Product Cost – Shipping cost) / Net Revenue	Ensures growth isn't coming at the cost of profitability.

Incentive	Discount Rate	Total Discount Amount / Gross Revenue	Monitors if Variant B relies too heavily on price cuts.
Experience	Median Time-to-Purchase	Median(Time _{purchase}) – Median(Time _{checkout})	Measures if the new UI reduces or increases user friction.

4. Scope & Eligibility

- **Definition:** Session_Platform = 'Web' AND User_Status = 'Logged-in' AND Eligibility_Flag = TRUE.
- **Variant Assignment:** Only eligible sessions are randomly assigned to **Variant A (Control)** or **Variant B (Treatment)**.
- **Baseline Monitoring:** Sessions that are Not-Eligible (e.g., Guest users or Mobile users) will be tracked as a "Natural Control."

Purpose: This allows us to verify that any changes seen in Variant B are actually due to the new UI and not just a site-wide trend (like a holiday sale) that affected everyone.

- **Data Quality:** Sessions with missing device data will be flagged. Revenue outliers are handled via IQR Capping to ensure the Mean Revenue is not skewed by extreme 'whale' purchases.

5. Stakeholder Questions & Data Strategy

- **Should we roll out B to everyone?**
 - *Strategy:* Conduct a Z-test for proportions on CVR (B vs. A). A "Go" requires $p < 0.05$ and a positive delta in RPES.
- **Where are the biggest funnel leaks?**
 - *Strategy:* Build a Comparative Funnel Drop-off Table. The step with the highest drop-off rate delta between A and B will be the focus for immediate optimization.
- **Which segments drive results?**

- *Strategy:* Perform Heterogeneous Treatment Effect (HTE) analysis across Marketing Channels and City Tiers to see if B over-performs in specific demographics.
- **Did the new experience confuse loyalists?**
 - *Strategy:* Segment CCR and Time-to-Purchase for **Returning Users**. If time increases significantly while conversion stays flat, the UI may be too jarring.
- **What is the revenue impact for next month?**
 - *Strategy:* Multiply the **Incremental RPES** (B minus A) by the projected 30-day session volume to provide Best/Base/Worst-case financial forecasts.